SAFMC 2026 — Autonomous High-Speed Drone Flock

Technical Summary | Tommy Lin (Project Manager & Lead Engineer)

Competition. Singapore Amazing Flying Machine Competition (SAFMC) 2026, Category High-Speed Drone Flock: a flock of 2–3 drones must autonomously fly a known indoor circuit ($\sim$40 m $\times$ 20 m play field), passing through a sequence of gates as quickly as possible while staying together as a group — no human control.

My role. Project Manager and Lead Engineer of the 9-person 2026 team, building on a framework inherited from our 2025 entry. My own hands-on contribution was the **path-planning algorithms and the simulation environment** — the role I held on the 2025 team — while as lead engineer I directed the full autonomy stack (all subsystems except the UWB sensor firmware, a teammate's work, which I integrated).

Status. The full autonomy stack is validated in simulation (Gazebo, full flock mission); the two-drone hardware platform was built and flight-tested. Closing the sim-to-real gap for the fully autonomous mission is the system's open frontier — and the origin of my current research direction.

System architecture

A ROS 2 (Humble) stack over a PX4 (v1.15) flight controller, organized as Perception → Decision → Control → Hardware:

- **Perception (vision).** A front camera detects ArUco markers (4 per gate). Gate pose is computed via PnP (`cv2.solvePnP`, IPPE_SQUARE) and stabilized with exponential-moving-average filtering.
- **Localization.** UWB ranging from 4 arena anchors (teammate's firmware) is fused with PX4 odometry; I integrated it as PX4 external odometry for global positioning.
- **Decision.** A hierarchical behavior state machine (Idle → Takeoff → Navigate → Pass-Gate → Wait-for-Flock → Land → Finish). Each behavior is a module behind a shared interface; transitions run at 10 Hz.
- **Control.** PX4 offboard mode using position (GotoSetpoint) and velocity (TrajectorySetpoint) commands; gate passing uses a proportional controller on the vision error to center the drone in real time.
- **Robustness.** Layered fallbacks — vision-based centering first, UWB-based gate-crossing as backup, and a distance/timeout failsafe to force progress.
- **Simulation.** A Gazebo arena (gates, drone, field) for validating the full stack end-to-end.

Key engineering challenges

- High-speed, precise gate passing under noisy perception → sensor fusion + EMA filtering + proportional control.
- Real-time coordination of perception, localization, decision, and control on a fixed 10 Hz loop.
- Graceful degradation when a sensor drops mid-flight → explicit multi-tier fallback logic.

Honest scope notes

- The base framework was inherited and evolved from the 2025 entry; my hands-on contribution was the path-planning and simulation work, plus PM / lead-engineer ownership of the 2026 stack.
- Perception is classical (ArUco markers), not learning-based; the UWB firmware was a teammate's.

Where I want to go next (research bridge)

This system hand-engineers its perception and high-level decisions. I want to replace those with **learning-based, language-conditioned** policies — VLA models for perception-to-action and agentic / LLM planning for high-level task decomposition — and to study decision-making under uncertainty, the focus of my current CS4246 coursework.